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Charting the unseen – A journey into disaster displacement risk

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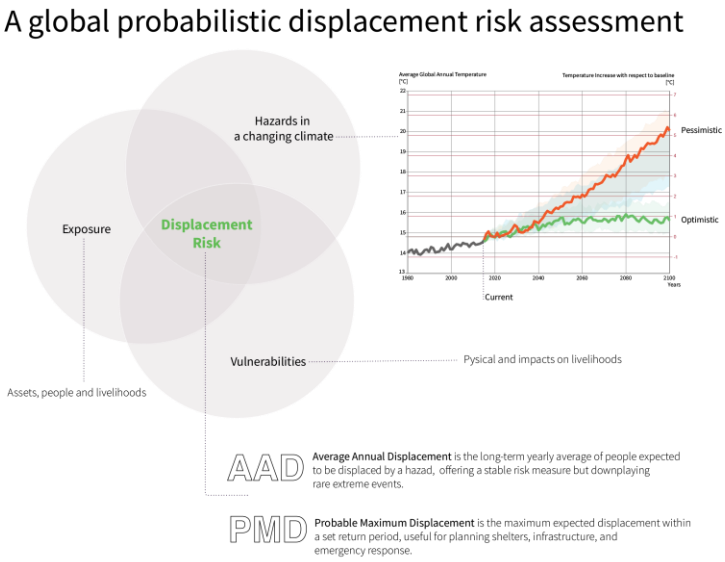
1 Introduction and objectives

Geophysical and weather-related hazards in a changing climate are driving unprecedented levels of human displacement. Disasters triggered 264.8 million internal displacements, or forced movements of people, across 210 countries and territories between 2015 and 2024. Quantifying this risk is essential for shaping adaptation strategies, policy development, and climate negotiations. This study introduces a globally consistent, modular risk modelling framework that assesses displacement risks from coastal and river flooding, droughts, earthquakes, and cyclonic winds, calibrated using historical disaster displacement events. The framework provides a unified, cross-hazard approach that enables evidence-based decision-making for anticipatory action and disaster risk reduction within climate informatics. Drawing on principles from catastrophe modelling in insurance catastrophe — traditionally used to estimate economic losses such as Average Annual Loss (AAL) and Probable Maximum Loss (PML)—this approach shifts the focus from the monetary value of residential buildings to the number of people who live in them, their livelihood dependencies, and their vulnerabilities to shocks.

2 Methods and data

The framework employs three event-based risk models: CLIMADA (cyclonic winds, coastal floods, droughts), CIMA (river floods), and GRM (earthquakes). All models generate

consistent outputs—Annual Average Displacement (AAD) and Probable Maximum Displacement (PMD) facilitating comparisons across hazards and regions. The models build upon the traditional risk analysis approach, whereby hazard, exposure, and vulnerability data are integrated calibrated using historical event-based data.



Hazard layers for river floods, coastal floods, droughts, earthquakes, and tropical cyclone winds were derived from state-of-the-art weather and geophysical impact models, combining climate projections, synthetic event catalogues, and scenario-based future pathways to capture event frequency and intensity. Exposure is characterized through the exploration of multiple global and open-source exposure layers to capture the full "riskscape" associated with all modeled hazards. These layers are combined with hazard- and region-specific vulnerability or impact functions that link hazard intensity to structural damage and, together with socioeconomic data, enable the estimation of housing and livelihood losses and, ultimately, displacement.

3 Preliminary results and/or expected outcomes

The magnitude of future displacement risk that countries may face is quantified using AAD. At the global scale, estimates range from approximately 15 million displacements from riverine flooding, followed by coastal flooding including storm surges (around 10 million), drought (4 million), earthquakes (3 million), and, lastly, cyclonic winds (around 167,000). Importantly, this metric does not represent the number of displacements that will occur in any given year. Rather, it reflects the average number of people that can be expected to be displaced annually when considering the full range of possible events over long timeframes. Our model results suggest that by 2100, weather-related displacement risk increases substantially under both optimistic and pessimistic scenarios, 75% and 157% respectively, evaluated using a present-day exposure layer to isolate changes in risk attributable to climate change. Additionally, the outputs identify regional hotspots and underscore the complexity of displacement drivers, including physical damage, livelihood loss, and income inequality. To better understand these dynamics, we complement the risk outputs with a multi-dimensional vulnerability assessment, exploring how environmental, economic, demographic, political, and social indicators interact across geographies to influence displacement risks. We combine human mobility theory and dimensionality reduction techniques to create an explainable statistical model of the drivers that accounts for the uncertainty and noise in displacement data. This study advances climate informatics by

providing a first-of-its-kind, multi-hazard displacement risk framework. These unified outputs enable holistic risk comparisons, supporting policy discussions on climate adaptation, Loss and Damage, and disaster risk reduction, and are well-suited for impact forecasting and anticipatory action.

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